

Updating parameters

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Apply a learning rate to an ordered gradient

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Look for: Estimate one partial derivative while other parameters stay fixed

Reference example, not the only acceptable wording: Teacher explanation: Prediction 3, residual 0, loss 0, gradient [0, 0].

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate per-example squared error and mean squared error

Reference example, not the only acceptable wording: Teacher explanation: Residuals are -1 and 2, squares 1 and 4, MSE 2.5.

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how one layer supplies the next layer inputs

Reference example, not the only acceptable wording: Teacher explanation: Scores are -2 and 2; activations are 0 and 2; $y = 0 - 2 + 1 = -1$.

Use two 45-minute sessions and keep old and new parameter boxes separate. This lesson checks one update, not a full optimiser. Treat the mixed-state procedure as different from the stated algorithm, rather than claiming all sequential methods are invalid.

Outcomes

Apply a learning rate to an ordered gradient

Update all parameters from the same saved state

Verify the new loss and identify overshooting

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The residual is -1 and the gradient is [-4, -2].

End goal for the pictorial case

By the end, I can explain updating parameters using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Gradient descent uses $\text{new parameter} = \text{old parameter} - \text{learning rate} \times \text{gradient entry}$. The minus sign chooses the locally downhill direction for a positive learning rate. A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: $x = 2$, target 3, $y = 2w + b$ and $L = (y - 3)^2$. At $[w, b] = [1, 0]$, the loss is 1 and gradient is [-4, -2]. With learning rate 0.1, new $w = 1 - 0.1 \times -4 = 1.4$ and new $b = 0 - 0.1 \times -2 = 0.2$.

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal learning rate.

Predict the rows: unchanged parameters and loss 1 at rate 0; [1.2, 0.1] and loss 0.25 at rate 0.05; [1.4, 0.2] and loss 0 at rate 0.1; [2.2, 0.6] and loss 4 at rate 0.3. The helper performs one step on this fixed example; it assumes numeric inputs and does not choose a learning rate for you.

Term · Meaning

Term: Learning rate

Meaning: A multiplier that controls the size of a gradient update

Term: Gradient descent

Meaning: Updating parameters in the direction opposite the gradient

Term: Simultaneous update

Meaning: Computing all new parameter values from the same old state

Term: Overshoot

Meaning: A step that moves too far and can increase the objective

Use one gradient for one parameter snapshot

Stage · Saved values · Action

Stage: Before the step

Saved values: w is 1 and b is 0

Action: Compute gradient [-4, -2]

Stage: Apply rate 0.1

Saved values: Use both old parameters

Action: Produce [1.4, 0.2]

Stage: After the step

Saved values: Use both new parameters

Action: Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Guided: [1.2, 0.1] gives prediction 2.5, residual -0.5 and loss 0.25. Exit: no. The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Use one gradient for one parameter snapshot

Stage · Saved values · Action

Stage: Before the step

Saved values: w is 1 and b is 0

Action: Compute gradient [-4, -2]

Stage: Apply rate 0.1

Saved values: Use both old parameters

Action: Produce [1.4, 0.2]

Stage: After the step

Saved values: Use both new parameters

Action: Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual

step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Updating parameters

Use one explicit gradient update and then recompute the loss.

The facts we will use

Model $y=2w+b$, target 5. Start $w=1$, $b=1$; learning rate 0.05. Initial $y=3$, error $y-\text{target}=-2$, squared loss 4.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update

squared error

before



after



Bar length shows the supplied value; the marked baseline is zero.

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for $[w, b] = [2 \times (-2) \times 2, 2 \times (-2)] = [-8, -4]$.



2. Update simultaneously

New $w = 1 - 0.05 \times (-8) = 1.4$; new $b = 1.2$.



3. Check new loss

New $y = 4$, residual -1 , loss 1 : lower than 4 .

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute gradient

What happens and why: Gradient for $[w, b] = [2 \times (-2) \times 2, 2 \times (-2)] = [-8, -4]$.

Step: 2. Update simultaneously

What happens and why: New $w = 1 - 0.05 \times (-8) = 1.4$; new $b = 1.2$.

Step: 3. Check new loss

What happens and why: New $y=4$, residual -1 , loss 1 : lower than 4 .

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Gradient descent subtracts learning rate times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original $[w,b]$ before updating either one. The small step brings the prediction closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new prediction and loss must be recalculated.

What this example does not establish

One successful toy update is not evidence of convergence for a real network or generalisation to unseen data.

Change one thing in the pictured case

Change: Set learning rate 0.2 from the same starting point.

Prediction: New $w=2.6$, $b=1.8$, $y=7$, squared loss 4 : no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve the Week 10 gradient and write the update rule with parentheses around negative entries.

Time or stage: 7 to 18 minutes

Teacher action: Calculate the rate 0.1 worked step and recompute the loss from the new parameters.

Time or stage: 18 to 32 minutes

Teacher action: Complete W1 and the guided rate 0.05 calculation.

Time or stage: 32 to 45 minutes

Teacher action: Run or trace all four code rows and complete W2.

Time or stage: Session 2 start to 8 minutes

Teacher action: Rebuild the old-state snapshot and compare it with the mixed-state example.

Time or stage: 8 to 20 minutes

Teacher action: Complete W3 and discuss zero gradient with W5.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including old loss, both new parameters and new loss.

Time or stage: 35 to 45 minutes

Teacher action: Use the exit response and a fresh retry. Keep the model-quality claim separate from implementation correctness.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by $(1 - 10 \text{ times rate})$. Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

Reduce help deliberately

Completion task: For the lesson start $[1, 0]$, apply rate 0.3 and calculate prediction and loss.

Teacher check: Teacher explanation: Gradient $[-4, -2]$ gives $[2.2, 0.6]$, prediction 5 and loss 4, larger than the starting loss 1.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Subtracting a negative adds a positive amount; show the signed calculation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

A gradient supplies a local direction, but step size can overshoot. Rate 0.2 gives the same loss 4 despite using the correct gradient. Changed-case check: New $w = 2.6$, $b = 1.8$, $y = 7$, loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Changed-case reasoning: New $w = 2.6$, $b = 1.8$, $y = 7$, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Subtracting a negative gradient decreases the parameter: Subtracting a negative adds a positive amount; show the signed calculation.

A downhill direction allows any rate: A finite step can overshoot and increase loss.

Recompute each gradient after its assignment: For this algorithm, all entries are computed at the same saved state.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: [1.2, 0.1] gives prediction 2.5, residual -0.5 and loss 0.25. Exit: no. The mixed-state procedure lowers this loss but does not implement the specified simultaneous gradient step; compare states and gradients, not just whether loss decreased.

W1 Apply a negative gradient

Starting with parameter $a = 2$, gradient -3 and learning rate 0.2, calculate the new a .

Teacher explanation: New $a = 2 - 0.2 \text{ times } -3 = 2.6$.

W2 Detect overshooting

For the lesson start [1, 0], apply rate 0.3 and calculate prediction and loss.

Teacher explanation: Gradient $[-4, -2]$ gives $[2.2, 0.6]$, prediction 5 and loss 4, larger than the starting loss 1.

W3 Preserve one snapshot

Explain why recomputing the bias gradient after changing w is not the simultaneous update specified here.

Teacher explanation: The two gradient entries then belong to different parameter vectors. Save the full old gradient and use it for both assignments.

W4 Update a new model independently

Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient $[2 \text{ times residual times } x, 2 \text{ times residual}]$, new parameters and new loss.

Teacher explanation: Prediction 2, residual -2, gradient $[-4, -4]$. New parameters $[1.4, 1.4]$ give prediction 2.8, residual -1.2 and loss 1.44, down from 4.

W5 Evaluate a zero gradient

At a point with gradient $[0, 0]$, what does this update do? Does that alone prove a global minimum on any loss surface?

Teacher explanation: It leaves parameters unchanged for a finite learning rate. Zero gradient alone does not establish a global minimum; stationary points can have other meanings.

Independent assessment and fresh retry

W4 requires gradient $[-4, -4]$, parameters $[1.4, 1.4]$ and loss 1.44. Fresh retry: $x = 1$, target 1, $w = 1$, $b = 1$, rate 0.25 gives residual 1, gradient $[2, 2]$, new parameters $[0.5, 0.5]$ and loss 0. Require both updates to use the original residual.

Assess independent transfer

Selected case: Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient $[2 \text{ times residual times } x, 2 \text{ times residual}]$, new parameters and new loss.

Selected case answer

Teacher explanation: Prediction 2, residual -2, gradient $[-4, -4]$. New parameters $[1.4, 1.4]$ give prediction 2.8, residual -1.2 and loss 1.44, down from 4.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled updating parameters evidence sheet to Image-pattern Network Investigation. Show how you can apply a learning rate to an ordered gradient. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use separate old, gradient and new columns. Extension: for the lesson example, show that the new residual is the old residual multiplied by $(1 - 10 \text{ times rate})$. Explain why rates 0 and 0.2 preserve loss while 0.3 raises it for a nonzero residual.

Week 12 repeats the training cycle for a small network.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/12, ai-5/14, ai-5/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.